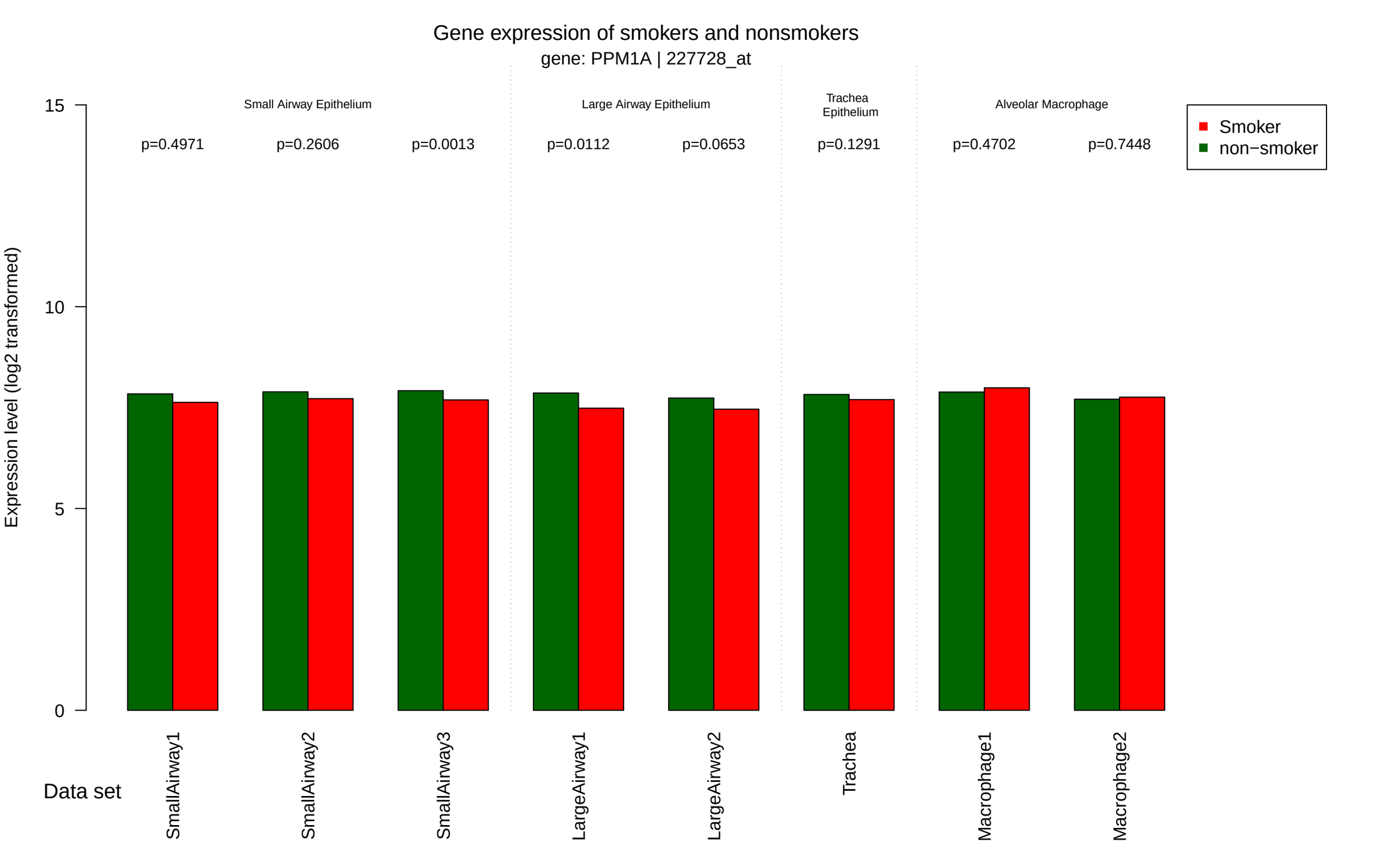
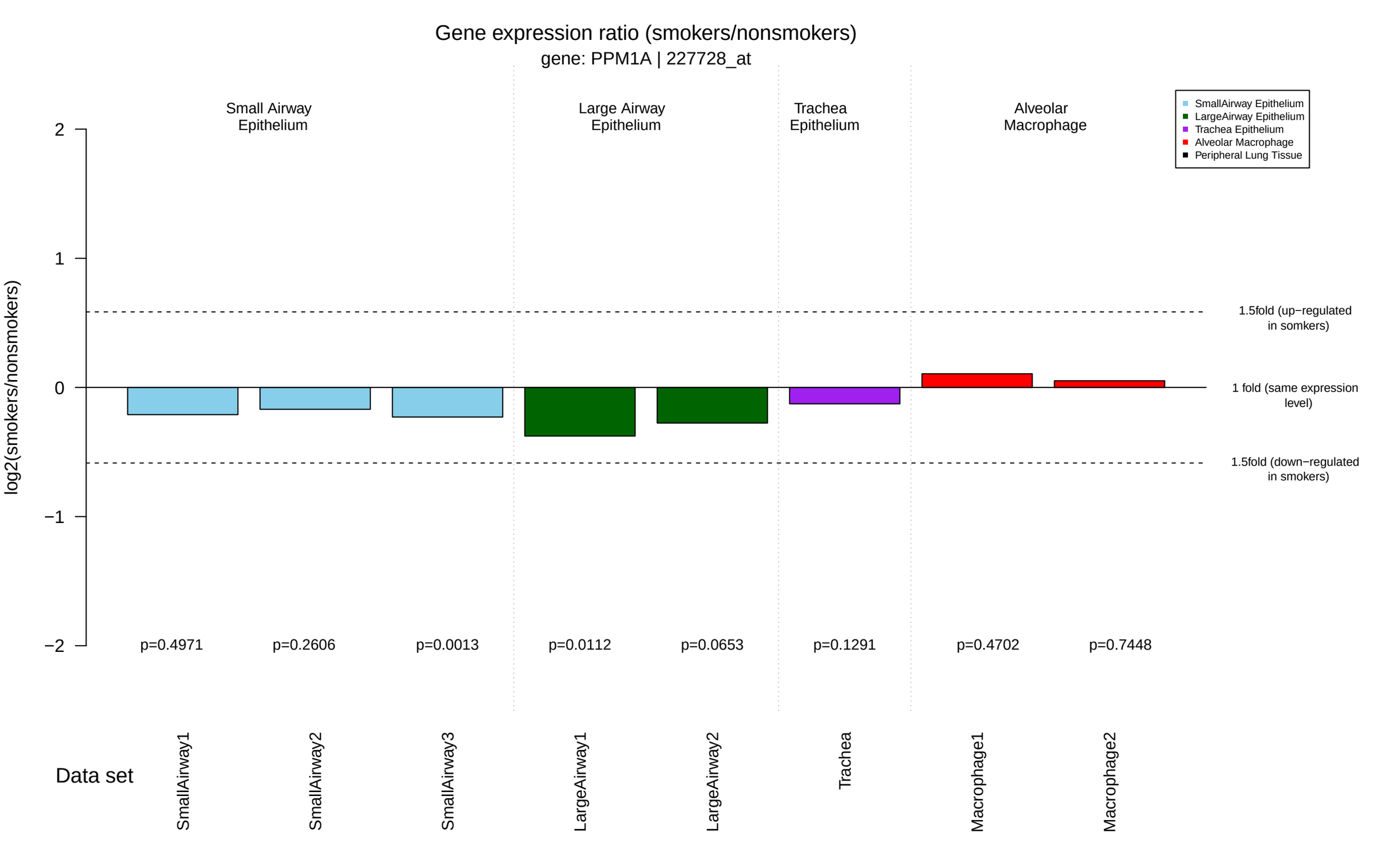
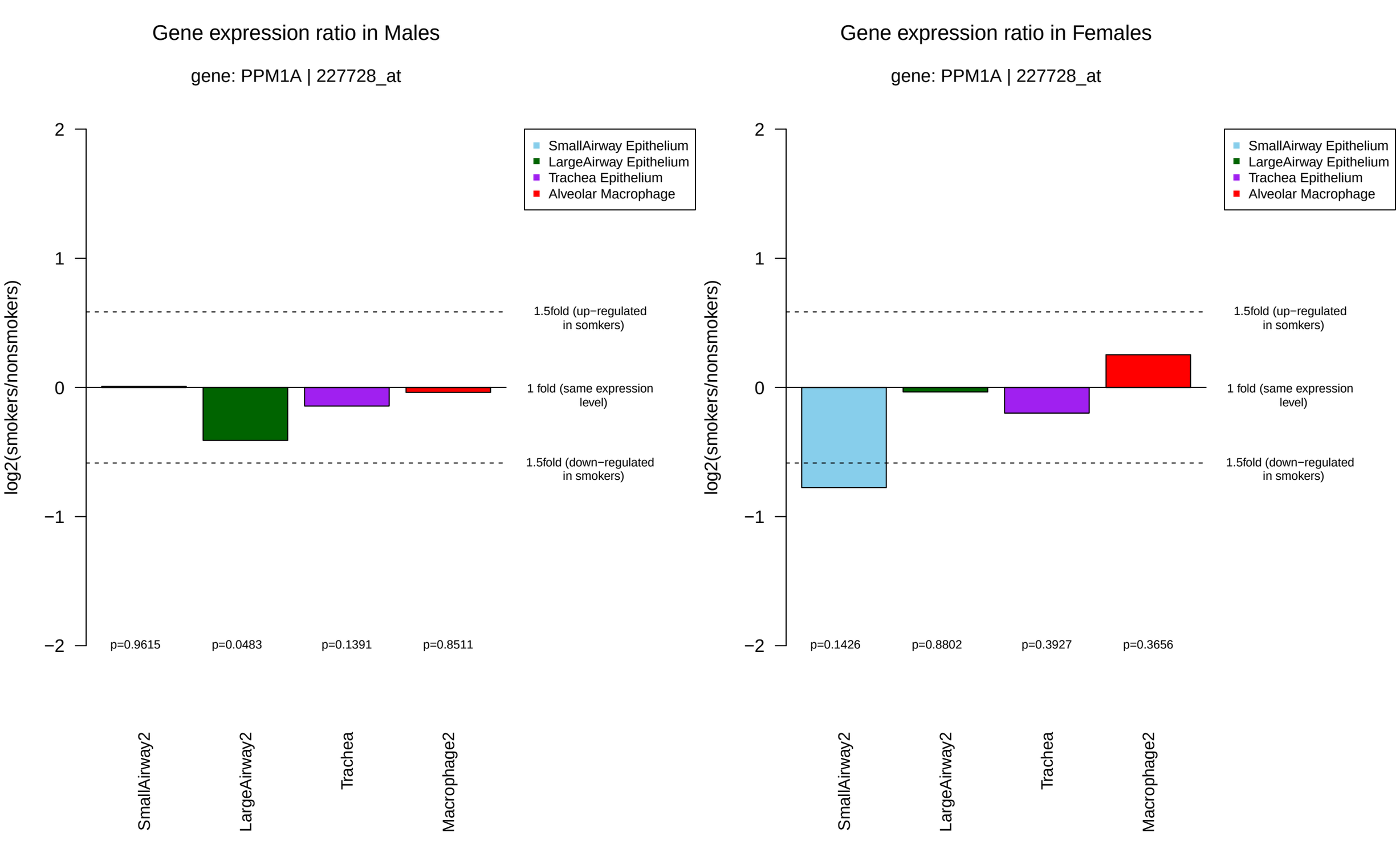
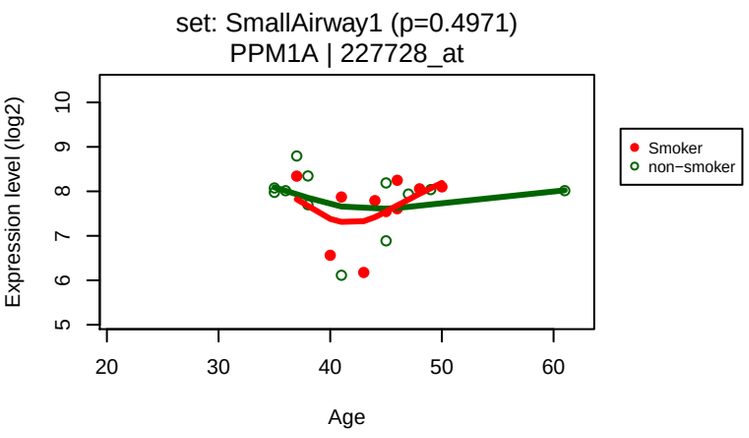
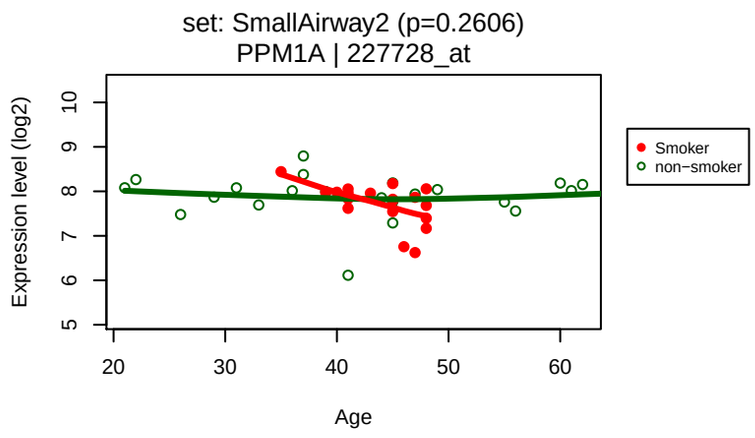
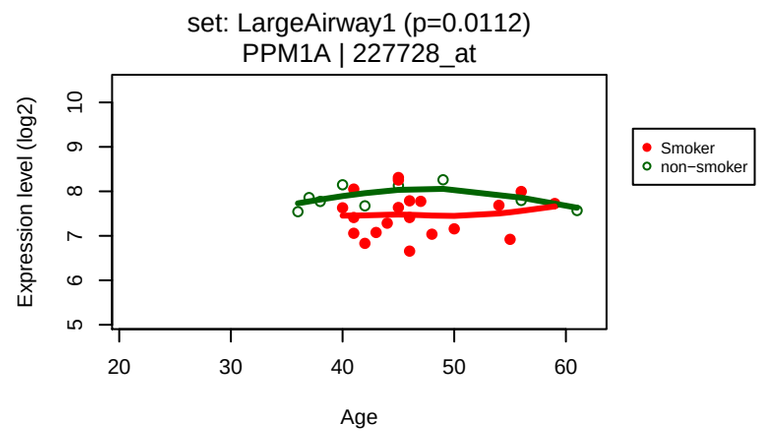

Gene expression of smokers and nonsmokers
gene: PPM1A | 227728_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4971
p=0.2606
p=0.0013
p=0.0112
p=0.0653
p=0.1291
p=0.4702
p=0.7448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

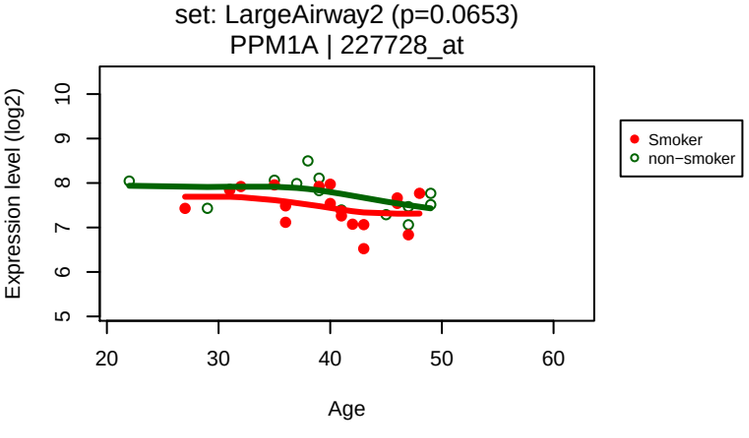

Gene expression ratio (smokers/nonsmokers)
gene: PPM1A | 227728_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4971
p=0.2606
p=0.0013
p=0.0112
p=0.0653
p=0.1291
p=0.4702
p=0.7448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

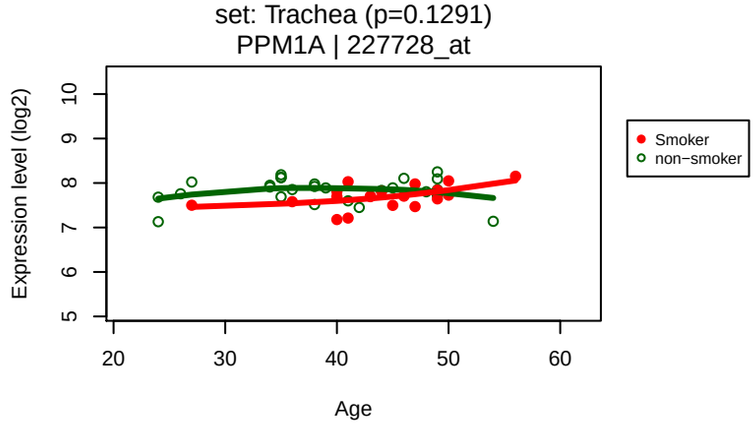

Gene expression ratio in Males
gene: PPM1A | 227728_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9615
p=0.0483
p=0.1391
p=0.8511
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1A | 227728_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1426
p=0.8802
p=0.3927
p=0.3656
SmallAirway2
LargeAirway2
Trachea
Macrophage2

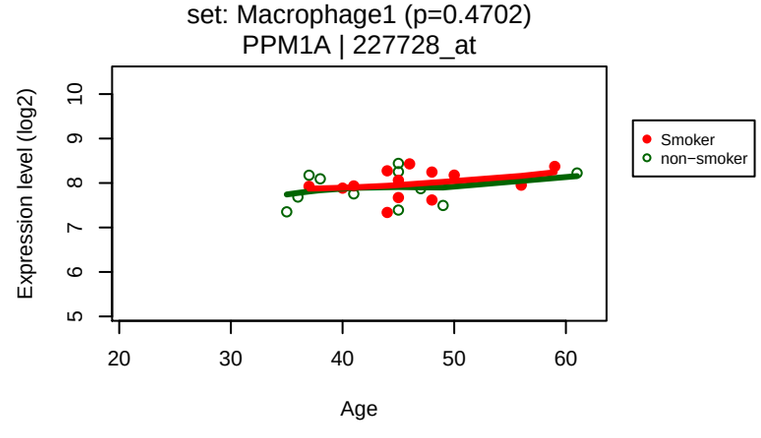

set: SmallAirway1 (p=0.4971)
PPM1A | 227728_at
Expression level (log2)
Age
Smoker
non−smoker

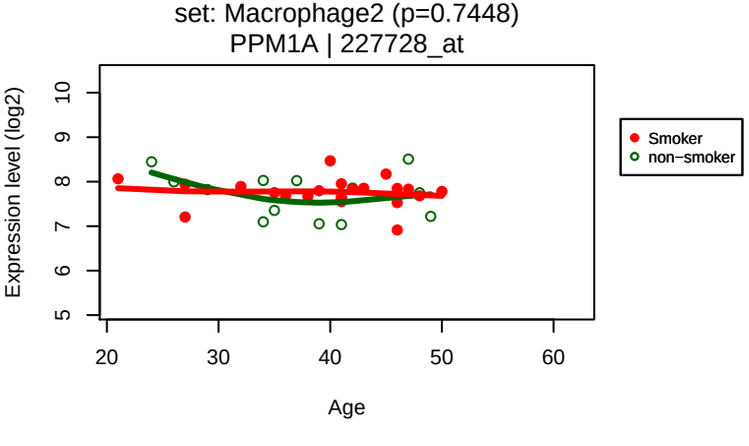

set: SmallAirway2 (p=0.2606)
PPM1A | 227728_at
Expression level (log2)
Age
Smoker
non−smoker

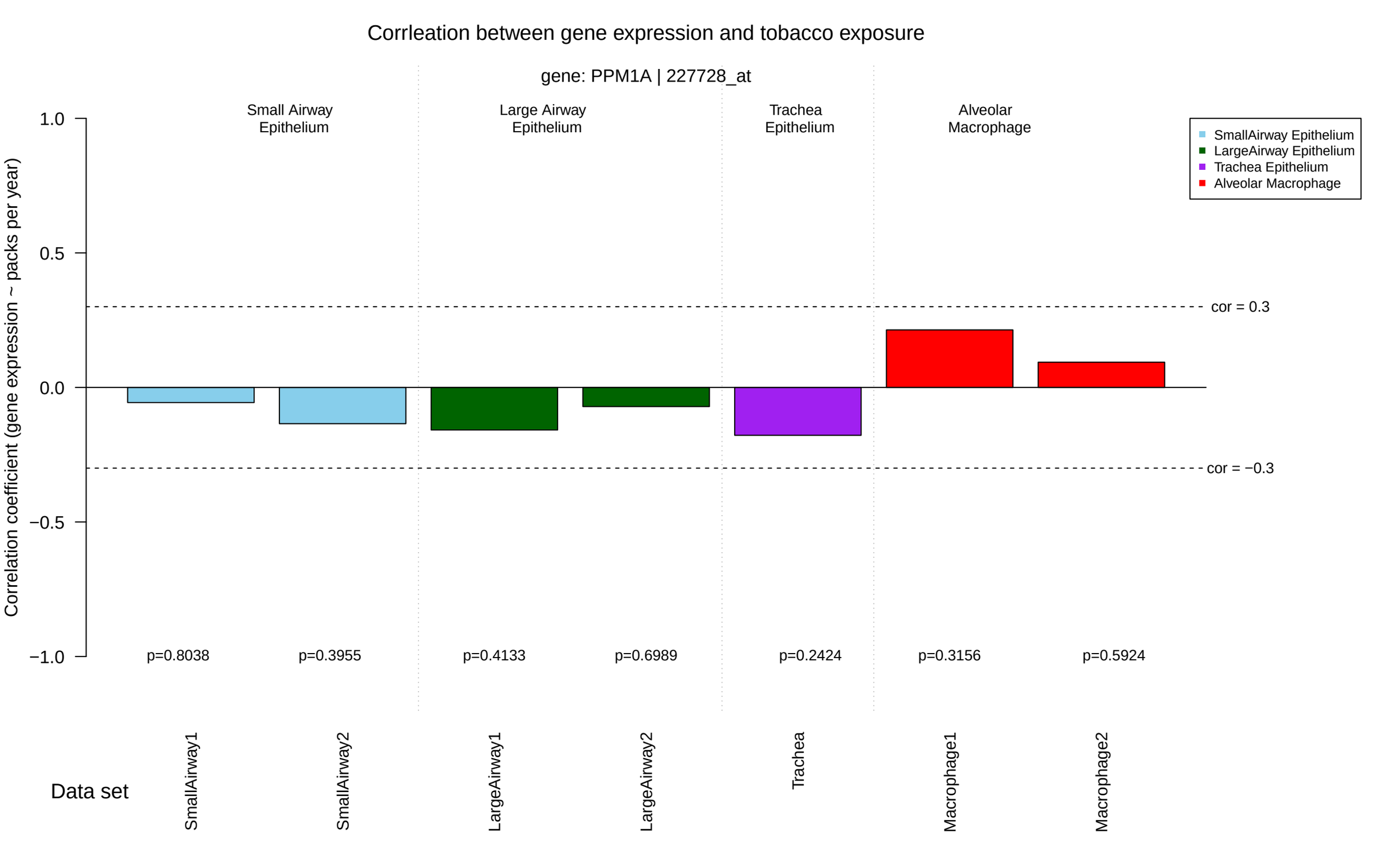

set: LargeAirway1 (p=0.0112)
PPM1A | 227728_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0653)
PPM1A | 227728_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1291)
PPM1A | 227728_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4702)
PPM1A | 227728_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7448)
PPM1A | 227728_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1A | 227728_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8038
p=0.3955
p=0.4133
p=0.6989
p=0.2424
p=0.3156
p=0.5924
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2